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Review

AIOps for log anomaly detection in the era of LLMs: A systematic literature review

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ABSTRACT

Modern IT systems generate large volumes of log data that challenge timely and effective anomaly detection. Traditional methods often require intensive feature engineering and struggle to adapt to dynamic operational environments. This Systematic Literature Review (SLR) analyzes how Artificial Intelligence for IT Operations (AIOps) benefits from advanced language models, emphasizing Large Language Models (LLMs) for more effective log anomaly detection. By comparing state-of-art frameworks with LLM-driven methods, this study reveals that prompt engineering – the practice of designing and refining inputs to AI models to produce accurate and useful outputs – and Retrieval Augmented Generation (RAG) boost accuracy and interpretability without extensive fine-tuning. Experimental findings demonstrate that LLM-based approaches significantly outperform traditional methods across evaluation metrics that include F1-score, precision, and recall. Furthermore, the integration of LLMs with RAG techniques has shown a strong adaptability to changing environments. The applicability of these methods also extends to the military industry. Consequently, the development of specialized LLM systems with RAG tailored for the military industry represents a promising research direction to improve operational effectiveness and responsiveness of defense systems.

1. Introduction

This study draws its foundations in the advancement of Artificial Intelligence for IT Operations (AIOps), a concept first introduced by Gartner in 2018 (Prasad & Rich, 2019). AIOps platforms leverage Machine Learning (ML) and deep learning algorithms to process and analyze vast volumes of operational data from multiple sources, automatically detecting and responding to system issues in real-time (Prasad & Rich, 2019). The adoption of these platforms is driven by the increasing complexity and scale of modern IT environments, which require advanced analytical capabilities beyond traditional methods to ensure service availability, optimize operational efficiency, and minimize downtime (Prasad & Rich, 2019). Recent investments by leading industry vendors such as IBM,[1] BigPanda,[2] ServiceNow,[3] Splunk,[4] Dynatrace,[5] and Datadog[6] underscore the importance and growing recognition of AIOps. Within this context, the emergence of Large Language Models (LLMs) has introduced novel opportunities to integrate AI-driven automation into production processes by enabling more efficient processing of vast amounts of unstructured data (Vitui & Chen, 2025).

This research primarily focuses on the use of LLMs in IT operations, assessing their efficacy in enhancing anomaly detection, which is one of the core tasks within AIOps. By anomaly, we define it as any abnormal log message within the internal logic of a system. Through a comprehensive analysis of current research, methods, and challenges, this Systematic Literature Review (SLR) aims to provide valuable insights into the evolving landscape of AIOps.

The evolution of AIOps marks a transformative shift in the management of complex IT systems. AIOps leverages advanced AI techniques to transform raw operational data (e.g., logs, metrics, incident reports)

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[1] IBM (International Business Machines Corporation) is a multinational technology company specializing in AI, cloud computing, and enterprise IT solutions. More information: https://www.ibm.com
[2] BigPanda provides AI-driven IT operations and incident automation solutions. More information: https://www.bigpanda.io
[3] ServiceNow offers AI-powered IT service management and workflow automation solutions. More information: https://www.servicenow.com
[4] Splunk provides data analytics and AI-powered monitoring solutions for IT and security operations. More information: https://www.splunk.com
[5] Dynatrace specializes in AI-driven observability and performance monitoring. More information: https://www.dynatrace.com
[6] Datadog offers cloud monitoring and security analytics powered by AI. More information: https://www.datadoghq.com

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into actionable insights that allow systems to detect, find root causes, and remediate anomalies proactively (Zhang, Jia et al., 2024). This approach improves the availability and reliability of services and reduces the need for manual intervention from IT professionals. In the end, this drives down operational costs and improves overall system efficiency (Notaro et al., 2021; Poenaru-Olaru et al., 2024).

The core tasks within AIOps revolve around data preprocessing (e.g., log parsing, metrics imputation, input summarization), failure perception (e.g., anomaly detection, failure prediction), root cause analysis (e.g., failure localization, failure category classification, root cause report generation), and automated remediation (e.g., assisted questioning, mitigation solution generation, command recommendations, script generation, automatic execution) (Zhang, Jia et al., 2024). Data preprocessing involves structuring operational data to enhance its usability for AI-driven analysis, particularly through log parsing and data filtering techniques (Zhang, Jia et al., 2024). Anomaly detection aims to identify abnormal patterns that indicate potential issues or failures (Zhang, Jia et al., 2024). Root cause analysis serves as a critical component in diagnosing failures, pinpointing their origins, and suggesting corrective actions (Zhang, Jia et al., 2024). Automated remediation, which represents the most advanced stage of AIOps, employs AI-driven decision-making to execute predefined mitigation strategies, reducing the need for manual intervention (Zhang, Jia et al., 2024).

The methodologies employed in AIOps are diverse, encompassing a range of AI and data-driven techniques. For anomaly detection, traditional ML, such as tree-based, clustering and PCA, RNNs, and GANs have long been used, typically relying on either supervised or unsupervised learning (He et al., 2017; Notaro et al., 2021; Vitui & Chen, 2025; Zhang, Jia et al., 2024). More recent advancements have integrated Natural Language Processing (NLP) to analyze log files and incident reports. Transformer-based architecture such as BERT[7] and GPT[8] have been leveraged for processing unstructured log data, improving anomaly detection significantly (Guan et al., 2024; Guo et al., 2024, 2021; Hadadi et al., 2024). A particularly innovative technique in AIOps has been the integration of LLMs into IT operations management. Unlike traditional models that require extensive feature engineering and retraining, LLMs have robust reasoning capabilities to interpret and extract meaningful insights from natural language data (Vitui & Chen, 2025; Zhang, Jia et al., 2024). LLMs have shown strong capabilities in interpreting unstructured data, making them highly effective for log analysis and system diagnosis (Vitui & Chen, 2025). In addition, LLMs have been trained on vast amounts of cross-platform data, which gives them a strong degree of generality (Zhang, Jia et al., 2024). Nonetheless, LLMs are still too general to be adaptive to multi-tasks, and companies exhibit significant limitations in developing these LLMs primarily due to data confidentiality (Chen et al., 2024). This is why the adoption of Retrieval-Augmented Generation (RAG), fine-tuning and prompt engineering techniques has increased recently (Chen et al., 2024; Zhang, Jia et al., 2024).

The integration of LLMs has also reached military and defense applications, where AIOps play a crucial role in ensuring cybersecurity, operational resilience, and decision-making (Loevenich et al., 2024). For instance, research shows that combining LLMs with reinforcement learning and rule-based systems enables autonomous cyber defense agents to analyze security breaches in real-time (Loevenich et al., 2024). Moreover, LLMs contribute to intelligence gathering and battlefield awareness as they are able to synthesize vast amounts of data from satellite imagery, communications, or sensor inputs, and this improves strategic planning (Cui & Gao, 2024).

The increasing digitization of military operations has resulted in the generation of large volumes of operational logs and telemetry data from systems such as radar, drones, communication equipment, and command-and-control platforms. As defense infrastructure becomes more reliant on complex IT ecosystems, the ability to detect anomalies – whether technical faults or signs of cyber intrusion – is becoming mission-critical. Traditional rule-based systems lack the adaptability and speed to manage these dynamic threats, necessitating the application of more intelligent, context-aware techniques such as LLMs combined with retrieval-based architectures.

The decision to emphasize the military domain in this study stems from the increasing strategic importance of AI-driven infrastructure within defense operations, where the stakes for anomaly detection are significantly higher than in conventional IT environments. In military contexts, log anomalies can indicate not only system failures but also potential cyberattacks, intelligence disruptions, or operational sabotage—events that can compromise mission-critical operations and national security. Unlike general IT systems, where anomalies may affect service delivery or performance, in military operations such disruptions can have immediate and irreversible consequences. Therefore, AIOps techniques and LLM-based anomaly detection solutions play a pivotal role in maintaining resilience, situational awareness, and proactive threat mitigation. That said, the methodological approach used in this review – particularly the inclusion criteria, paper selection, and analysis – was designed with a focus on military applications but is not limited to them. The identified patterns, frameworks, and insights are applicable across a wide range of high-stakes industries, such as finance, healthcare, and critical infrastructure, where reliability and anomaly detection are equally essential.

This study explores the use of LLMs for log anomaly detection and investigates the usefulness of the implementation of RAG to improve the specialization of LLMs. Thus, this literature aims to answer the following knowledge questions:

1. What are the main challenges and benefits of implementing AIOps in IT systems?
2. What AIOps techniques are being used for automating anomaly detection in IT systems?
3. What are the main evaluation metrics used for log anomaly detection using LLMs?
4. How does RAG improve the knowledge and reasoning capabilities of LLMs?

The insights gained from this systematic literature review explain the current state of research and gaps in the literature to bring future advancements in the field of AIOps. To the best of our knowledge, this is the first study that systematically investigates the intersection of AIOps, LLM-based log anomaly detection, and RAG integration. While prior works have primarily focused on traditional anomaly detection methods, such as statistical techniques, rule-based systems, or classical machine learning models, these approaches often suffer from scalability issues, limited contextual awareness, and reduced adaptability to new types of anomalies (Gao et al., 2023; He et al., 2017; Wang et al., 2024; Xu & Ding, 2024).

In contrast, our review highlights how LLMs, especially when augmented with RAG, offer a more robust and dynamic foundation for handling unstructured and semantically complex log data. This work contributes novel insights into how RAG can compensate for LLM limitations, such as hallucinations or domain drift, by incorporating external, up-to-date, and task-specific knowledge at inference time. By bridging these research areas, our study offers a comprehensive perspective that advances the understanding of next-generation anomaly detection frameworks within the broader AIOps ecosystem.

[7] BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained language model developed by Google that uses deep bidirectional transformers to understand the context of words in a sentence. It serves as the foundation for many state-of-the-art NLP tasks. More information: https://arxiv.org/pdf/1810.04805

[8] GPT (Generative Pre-trained Transformer) is a series of autoregressive language models developed by OpenAI that generate human-like text by predicting the next token in a sequence. The models are pre-trained on large text corpora and fine-tuned for various natural language processing tasks. More information: https://openai.com/research/gpt

Table 1
Article selection criteria.

Criteria	Decision
Pre-defined keywords list	Inclusion
Document type: conference paper, article	Inclusion
Subject area: computer science, engineering	Inclusion
Article written in English	Inclusion
Article published before 2021[a]	Exclusion
Unavailability of free article version	Exclusion

[a] For the initial inclusion/exclusion criteria in the Scopus database, only articles published from 2021 onwards were included. However, through the snowball process, additional articles published before 2021 were collected.

This paper is structured in the following way: Section 2 discusses the methodology developed to carry out this review, explaining the steps to extract relevant articles. Section 3 deconstructs the research landscape to identify trends, academic contributions, and emerging themes in AIOps research. Section 4.1 focuses on the current techniques, challenges, and benefits of AIOps. Section 4.2 addresses the use of LLMs in anomaly detection, and Section 4.3 explains what RAG is and how it can be implemented to improve reasoning capabilities in LLMs. Section 4.4 describes some recommendations for future work in the field of log anomaly detection and proposes a framework for log anomaly detection using LLMs and RAG. Finally, Section 5 concludes the paper.

2. Methodology

In this section, we detail the methodology used for conducting the systematic literature review. We describe the paper selection process, including inclusion and exclusion criteria, as well as the search queries and academic databases employed to ensure comprehensive coverage of the relevant literature.

The methodology employed in this study follows a structured approach to conducting a literature review on AIOps, the use of LLMs for anomaly detection, and the implementation of RAG to ensure unbiased outcomes (Amato et al., 2024; Firmansyah et al., 2024).

First, we established a set of knowledge questions that the literature review aims to answer (see Section 1). These questions served as the foundation of the research to ensure that the study remained focused on the targeted areas. Following this, we conducted a comprehensive exploration and analysis of the existing literature to gain a broader perspective on the field of AIOps. The goal was to map out the landscape of current research, identifying common methodologies, data-driven techniques, and evaluation methods to highlight emerging trends. During this phase, we defined inclusion and exclusion criteria (see Table 1) to filter out studies that did not meet certain standards. We selected studies based on parameters such as publication year, document type, language, pre-defined keywords list, and unavailability of a free article version. Next, we assessed the quality of the collected papers by reading their titles and abstracts, evaluating their credibility based on the rigor of the research rather than speculative ideas. Subsequently, we extracted data and compiled a table (see Appendix) with relevant information about each paper such as topic, professional setting, main purpose, data-driven techniques used, and evaluation metrics. Lastly, we summarized and discussed the most important outcomes of each paper in this review. Throughout the entire process, we ensured rigor, reliability, and reproducibility of the methodology, following other SLRs presented in the literature (Amato et al., 2024; Firmansyah et al., 2024; Notaro et al., 2021).

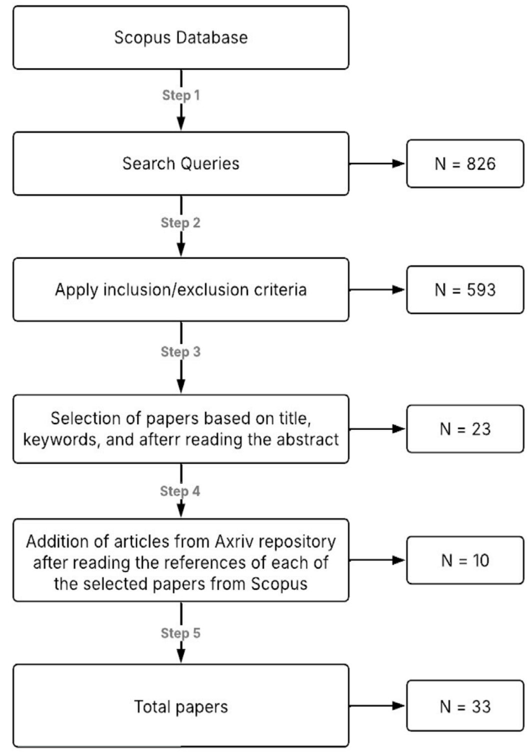

Fig. 1. Selection process of articles.

Scopus[9] was selected as the primary database, and arXiv[10] repository was used when some references of the papers were relevant to add in the literature review.[11] The advanced search feature was used to create custom queries to focus on the applications of LLMs with a base knowledge in log anomaly detection. Important keywords included “AIOps”, “anomaly detection”, “large language models”, and “retrieval-augmented generation”, in combination with “AND” and “OR” operators. The final search queries used were:

- (“artificial intelligence for IT operations” OR “AIOps”) OR (“incident detection” OR “anomaly detection”) AND (“large language models” OR “LLMs”) AND (“military” OR “transport” OR “ship”)
- (“artificial intelligence for IT operations” OR “AIOps”) AND (“incident detection” OR “anomaly detection”)
- (“LLMs” OR “large language models”) AND (“RAG” OR “rag” OR “retrieval-augmented generation”) AND (“anomaly detection” OR “incident detection”)

Fig. 1 shows a total of 33 articles gathered through the five-step strategy. We retrieved 23 articles by the search queries in Scopus. The manual inclusion followed a snowballing process (Wohlin, 2014) from references of the articles gathered.

[9] Scopus is a comprehensive bibliographic database that indexes peer-reviewed literature across multiple disciplines, including AI, technology, and computer science. More information: https://www.scopus.com

[10] arXiv is an open-access repository of research articles, particularly in computer science, physics, and AI, providing preprints before formal peer review. More information: https://arxiv.org

[11] While Scopus and arXiv provide extensive coverage of peer-reviewed and preprint literature, they are not entirely exhaustive. Their scope may omit relevant studies from smaller or non-indexed venues, and the diversity and completeness of indexed content can vary across subfields, potentially introducing selection bias.

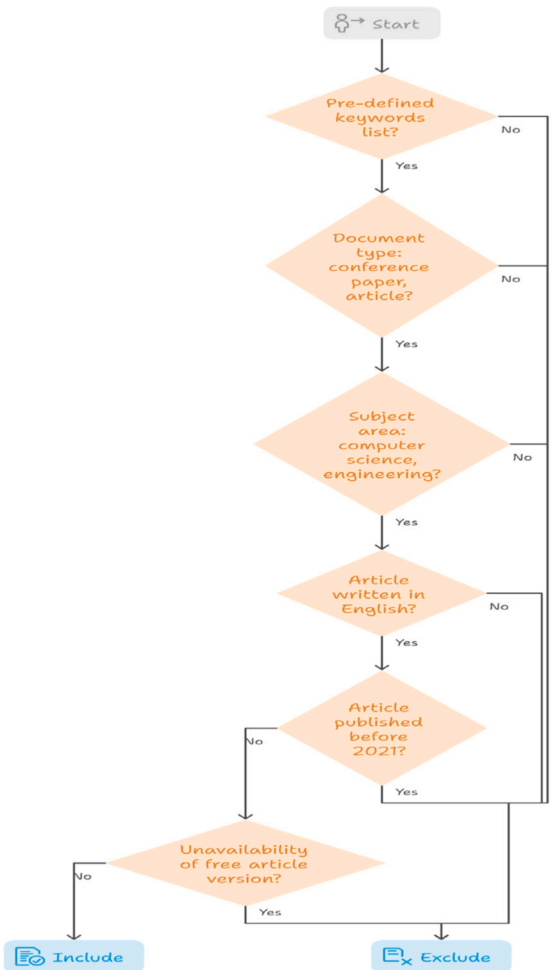

Fig. 2. Selection process: Inclusion and exclusion of papers.

Table 1 and Fig. 2 summarize the inclusion and exclusion criteria used in this study to ensure a rigorous and systematic selection of relevant literature. We included only publications that matched a pre-defined list of keywords, were classified as conference papers or journal articles, and belonged to the subject areas of computer science and engineering. To maintain consistency and interpretability, only articles written in English were considered. Furthermore, for the initial search in the Scopus database, we limited the selection to articles published from 2021 onwards to capture the most recent developments in the field; however, earlier publications were included through a backward and forward snowballing process when deemed relevant. Finally, we excluded papers for which no freely accessible version was available, as this would limit reproducibility and independent verification of the review process.

3. Deconstructing the research landscape

This section presents a structured analysis of the research landscape surrounding the applications of AIOps techniques for anomaly detection in three parts. First, we provide an overview of the distribution of articles over recent years. Second, we categorize the reviewed articles according to their research domain. Understanding this distribution helps determine which platforms are actively exploring AIOps and where new approaches are emerging. Last, we delve into the recurring keywords that define the current state of AIOps research within the collected articles. It is also important to highlight that the Appendix shows an overview of each article, ensuring transparency and clarity in the selection and evaluation of relevant articles.

3.1. Temporal distribution of articles

We start by analyzing the chronological timeline of articles in AIOps, RAG, AI in military, and anomaly detection. By tracking publication trends from 2021 to 2025, this analysis highlights periods of rapid growth in certain fields to better understand how the focus on AIOps research has adapted to new trends and technological advancements. Fig. 3 illustrates the increase in the number of articles published in recent years, particularly in the fields of RAG and anomaly detection. The rise in RAG-related articles reflects a growing interest in leveraging LLMs with external knowledge to mitigate hallucinations and improve performance in domain-specific tasks. AIOps articles have maintained a stable publication rate over the past years, which suggests that it still remains a significant area of interest in the software industry. In addition, the implementation of AI has been increasing in the defense and military industries in the past year, despite challenges related to classified information, indicating expanding roles for AI in operational domains such as image recognition and decision-making.

3.2. Journal distribution of articles

The analysis focused on a single table (see Table 2), which included the topic, author/s, journal of publication, number of citations, and Field-Weighted Citation Impact (FWCI). Understanding this distribution helped identify the platforms that actively explored AIOps and revealed where new approaches emerged. This information was important for assessing the quality and academic impact of each article, as it provided insight into the credibility and reach of the publications.

Table 2 summarizes the journal distribution across all selected articles, covering topics such as AIOps, anomaly detection, RAG, and AI applications in the military. The works by Notaro et al. (2021) and Dang et al. (2019) stand out with the highest number of citations within AIOps, suggesting their strong influence in the field. Similarly, He et al. (2017), Guo et al. (2021), and Zhu et al. (2018) received notable citations in the anomaly detection category, while Fan et al. (2024) was the most cited work in RAG-related research. The analysis also revealed that all articles were published in different journals, highlighting the diverse range of sources contributing to these topics.

Since the AIOps, and RAG with LLMs domains are relatively new and emerging topics, it is understandable that many studies are available as preprints in arXiv, lacking peer-review (Gao et al., 2023; Guan et al., 2024; Hadadi et al., 2024; Prasad & Rich, 2019; Su et al., 2024; Vitui & Chen, 2025; Xu & Ding, 2024; Zhang, Jiang et al., 2024). Furthermore, we can also notice from Table 2 that most of these articles are from the last two years, which explains why they do not yet have many citations, as they typically accumulate over time as research gains visibility within the community.

3.3. Keywords distribution across articles

Keywords analysis provides a clear vision of the main theme and the predominant topic and allows for a better understanding of the research landscape, facilitating future studies and supporting the identification of gaps within the literature.

Figs. 4 and 5 illustrate the keyword network graphs built for this study. These graphs show relationships between key terms extracted from the articles, providing insights into predominant themes and their connections over time. We focused on building keyword graphs across all articles, instead of concentrating on individual topics, to get an overall picture of the research landscape. We used VOSviewer[12] to

[12] VOSviewer is a software tool for constructing and visualizing bibliometric networks, commonly used for analyzing citation networks and co-authorship relations. More information: https://www.vosviewer.com

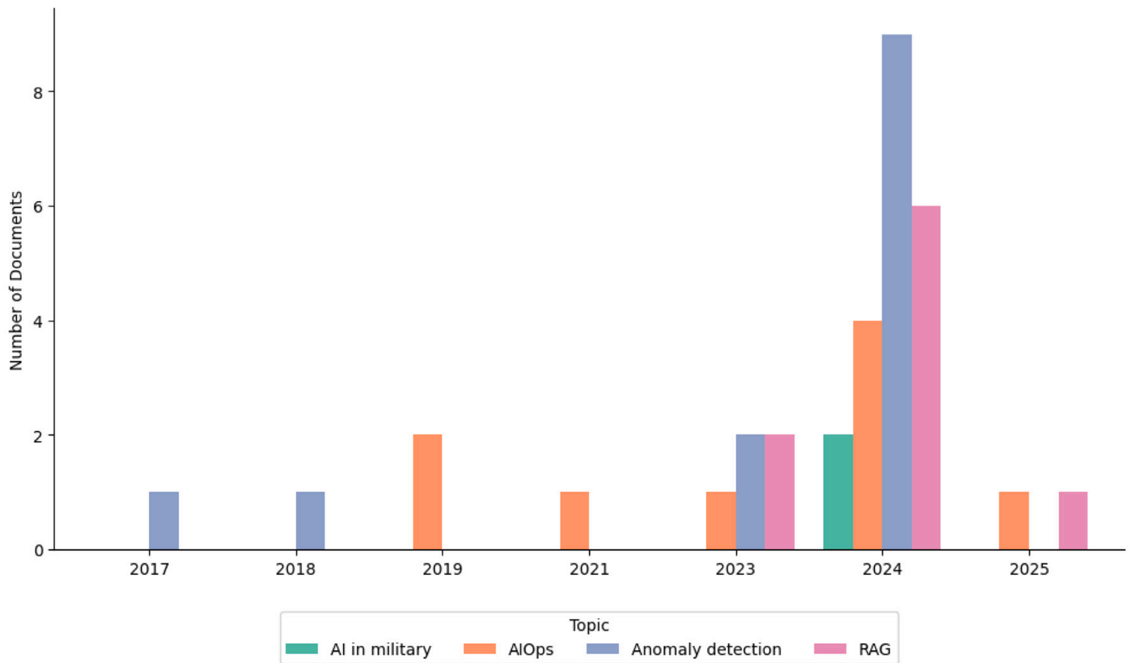

Fig. 3. Yearly distribution of collected papers.

Table 2
Journal distribution of articles.

Topic	Study	Journal/Conference	# Citations	FWCI
AIOps	Zhang, Jia et al. (2024)	Preprint on arXiv	–	–
AIOps	Poenaru-Olaru et al. (2024)	2024 IEEE/ACM CAIN Conference	1	–
AIOps	Notaro et al. (2021)	ACM Trans. on Intelligent Systems	49	3.30
AIOps	Diaz-De-Arcaya et al. (2023)	ACM Computing Surveys	18	10.18
AIOps	Mansour et al. (2024)	Int. J. of Engineering Trends	–	–
AIOps	Prasad and Rich (2019)	Gartner	21	–
AIOps	Duan et al. (2024)	Electronics (Switzerland)	–	–
AIOps	Vitui and Chen (2025)	Preprint on arXiv	–	–
AIOps	Dang et al. (2019)	2019 IEEE/ACM ICSE-Companion	160	17.03
Anomaly Detection	Wang et al. (2024)	IEEE Int. Conf. on Software Eng.	–	–
Anomaly Detection	Su et al. (2024)	Preprint on arXiv	–	–
Anomaly Detection	No et al. (2024)	Eng. Applications of AI	2	1.01
Anomaly Detection	Guo et al. (2021)	Proc. Int. Joint Conf. on Neural Networks	171	16.54
Anomaly Detection	Guan et al. (2024)	Preprint on arXiv	–	–
Anomaly Detection	He et al. (2017)	IEEE ICWS 2017	586	19.24
Anomaly Detection	Zhu et al. (2018)	IEEE ICSE-SEIP 2019	382	44.98
Anomaly Detection	Guo et al. (2023)	12th Int. Conf. on Learning Representations	4	7.61
Anomaly Detection	Mehrabi et al. (2024)	IEEE ICSME 2024	–	–
Anomaly Detection	Guo et al. (2024)	AAAI Conf. on AI	9	20.04
Anomaly Detection	Hadadi et al. (2024)	Preprint on arXiv	–	–
Anomaly Detection	Xu and Ding (2024)	Preprint on arXiv	–	–
Anomaly Detection	Liu et al. (2023)	IEEE Int. Conf. on Program Comprehension	3	5.74
Anomaly Detection	Almodovar et al. (2024)	IEEE Trans. Netw. Serv. Manag.	15	7.33
RAG	Vizniuk et al. (2024)	J. of AI and Soft Comp. Research	–	–
RAG	Arslan et al. (2024)	Procedia Computer Science	–	–
RAG	Fan et al. (2024)	ACM SIGKDD Int. Conf. on KDD	27	63.57
RAG	Jeon et al. (2025)	J. of Manufacturing Systems	–	–
RAG	Chen et al. (2024)	Proceedings of Science	–	–
RAG	Gao et al. (2023)	Preprint on arXiv	–	–
RAG	Zhang, Jiang et al. (2024)	EMNLP 2024 - Industry Track	–	–
RAG	Ovadia et al. (2023)	EMNLP 2024 - Main Conference	–	–
AI in Military	Cui and Gao (2024)	Lecture Notes in Electrical Eng.	–	–
AI in Military	Loevenich et al. (2024)	IEEE MILCOM	–	–

create the graphs. First, we constructed a network graph to visualize the relationships and connections between keywords, organizing them into clusters. Then, we overlaid an additional graph on top of the network graph to incorporate publication years to identify how research interests have evolved over time and how they can be connected with each other. The analysis unveiled "large language model", "anomaly detection", "AIOps", and "retrieval-augmented generation" as the recurring themes within the literature, followed by other interesting keywords such as "It operations", "log analysis", "fine-tuning", and "prompt engineering".

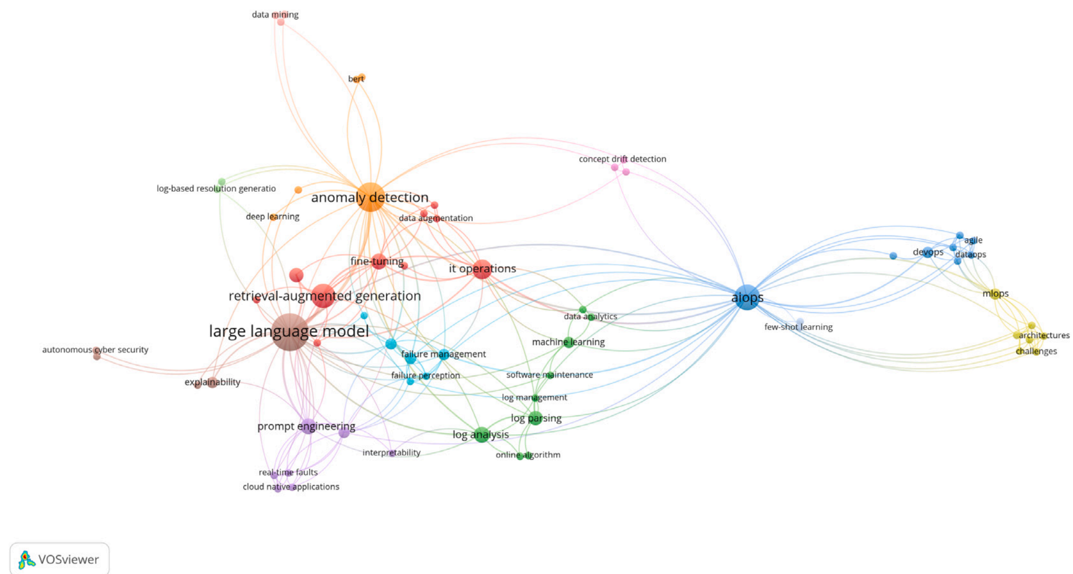

Fig. 4. Keyword network across all articles.

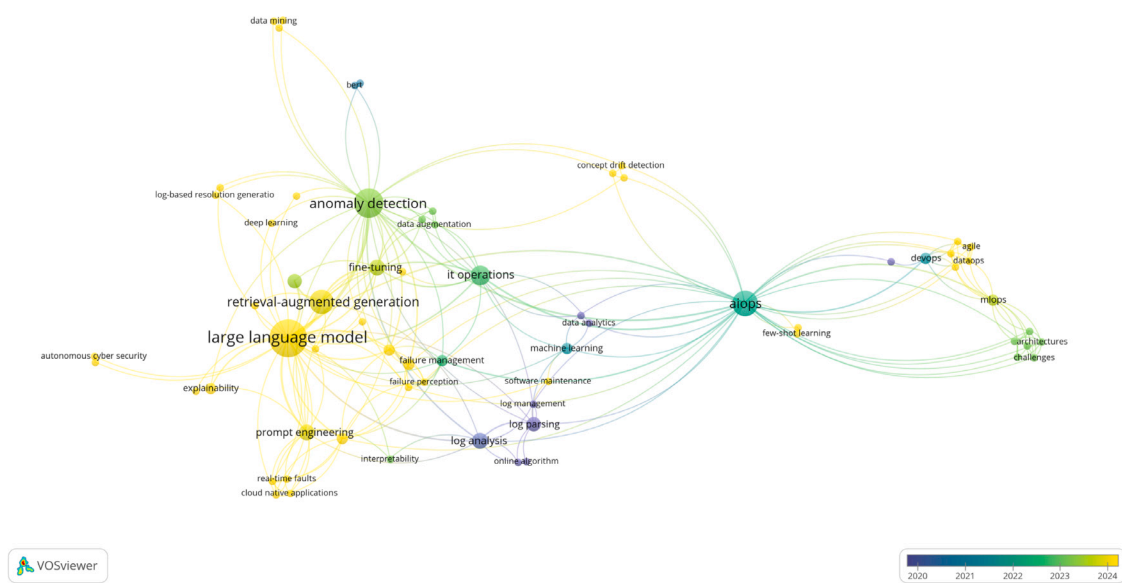

Fig. 5. Keyword overlay timeline.

Keyword networks revealed that research in recent years intersects around LLMs and anomaly detection within the AIOps domain, suggesting a shift from traditional models toward advanced NLP techniques in IT operations.

4. Qualitative summary and analysis of the literature

This section synthesizes the insights obtained from the reviewed studies on AI-driven operations (AIOps) and their role in log anomaly detection. We first discuss core AIOps techniques, associated challenges, and emerging trends. Next, we examine the application of Large Language Models (LLMs) in log anomaly detection, followed by a review of Retrieval-Augmented Generation (RAG) as a complementary approach for context-aware anomaly detection. Finally, we propose a theoretical framework and highlight promising directions for future research and applications.

4.1. *AIOps: Techniques, challenges, benefits, and future trends*

This section provides an overview of AIOps, focusing on the key techniques used in practice, the challenges faced during implementation, and the associated benefits. We also highlight emerging trends and future directions shaping the evolution of AIOps in modern IT systems.

AIOps represents an innovative approach to managing and monitoring new IT environments. By leveraging machine learning, big data analytics, and visualization techniques, AIOps platforms are designed to enhance efficiency and reliability in IT systems, providing operational insights that were not feasible with traditional methods (Prasad & Rich, 2019). The cycle follows a continuous three-stage process that observes data, engages with it to find patterns and root causes, and acts to resolve issues (Prasad & Rich, 2019). These capabilities not only improve customer satisfaction but also allow companies to shift towards more predictive approaches, instead of traditional management methods, reducing operational costs and downtime of IT services (Dang et al., 2019).

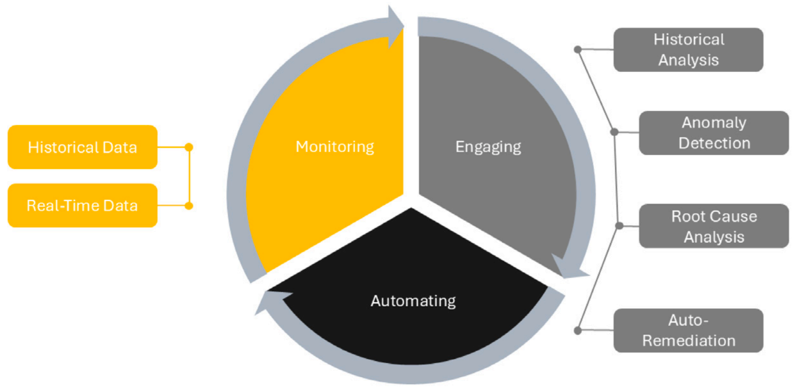

Fig. 6. AIOps operational cycle (Prasad & Rich, 2019).

At its core, AIOps integrates large amounts of operational data, from logs, metrics, traces, alerts, incident reports, and Q&A, to facilitate data preprocessing, failure perception, root cause analysis, failure prediction, and auto-remediation. These tasks are sequentially related, meaning that failure perception is based on preprocessed data, and once the anomaly has been identified in the system, a root cause analysis is performed to trace the anomaly and, in the end, appropriate auto remediation methods are implemented to mitigate the issue from the system (Zhang, Jia et al., 2024)), completing the operational cycle (see Fig. 6).

This systematic literature review mainly focuses on anomaly detection within AIOps because it precedes subsequent tasks. Emphasizing the initial techniques within AIOps allows for a more structured exploration of its processes. The preprocessing phase involves tasks such as log parsing for extracting meaningful information from logs, metrics imputation to estimate incomplete or missing data, and input summarization to highlight the most important information in the prompt (Zhang, Jia et al., 2024). Once the data is preprocessed, the datasets are ready to enable an effective failure perception, which involves predicting failures or detecting anomalies from historical trends (Zhang, Jia et al., 2024). Anomaly detection, as one of the core tasks in AIOps, is lately using advanced machine learning and deep learning techniques. Previous work has focused on unsupervised or semi-supervised models, but the best-performing and most popular techniques to detect anomalies in univariate data belong to signal reconstruction models (Poenaru-Olaru et al., 2024). Statistical methods such as Spectral Residuals (SR), Fast Fourier Transform (FFT), and Prediction Confidence Interval (PCI) identify strange behaviors in systems by encoding the time series into a latent space (Poenaru-Olaru et al., 2024). These techniques have low computational costs, but lose information during the encoding process (Poenaru-Olaru et al., 2024). In addition, to preserve more information, deep learning methods such as autoencoders, LSTMs, DONUT, and CNNs have proven strong performance when detecting anomalies (Notaro et al., 2021; Poenaru-Olaru et al., 2024).

Despite these advancements, real-world implementation of AIOps still faces significant challenges. Poenaru-Olaru et al. (2024) introduce the phenomenon of concept drift. Concept drift occurs when the properties of the incoming data change over time, meaning that the patterns the model originally learned no longer fully represent the current reality. For example, a system log model trained on older software versions may fail to detect anomalies once the software is updated and generates new types of logs. If models are not periodically retrained, this drift causes their performance to degrade. Other major challenges of AIOps include the complexity of getting high-quality and quantity labeled datasets, a difficult mindset shift towards new technologies, a gap in innovation that could guide people in different disciplines to build AIOps solutions, and limitations in model adaptability to specific tasks and across different software systems, as integration of AIOps with existing systems may be challenging (Dang et al., 2019; Zhang, Jia et al., 2024).

Conversely, adopting AIOps provides numerous substantial benefits. These include improved service quality and reliability, increased customer satisfaction, enhanced productivity of engineering teams, and significant reductions in operational costs due to proactive issue detection and management (Dang et al., 2019). Additionally, AIOps fosters improved collaboration among traditionally siloed IT teams, creating more streamlined operations and faster incident resolution (Mansour et al., 2024). Moreover, AIOps contributes to better scalability and agility in managing IT infrastructures by automating routine processes and minimizing complexities associated with managing large-scale systems (Notaro et al., 2021). Real-time data analytics and operational insights further allow rapid detection, diagnosis, and resolution of performance bottlenecks, enhancing overall operational efficiency (Prasad & Rich, 2019).

A recent study from Duan et al. (2024) shows that meta-learning has been introduced to further enhance the adaptability and effectiveness of anomaly detection. Meta-learning involves using prior knowledge to enable AI to autonomously learn new tasks, with the goal of making a model generalizable by 'learning how to learn'. Typically, there are few examples of specific types of anomalies, making traditional learning methods less effective (Duan et al., 2024). To address this issue, Duan et al. (2024) apply the Model-Agnostic-Meta-Learning (MAML) framework to facilitate the model's ability to quickly update its parameters when encountering new anomalies. The experimental results demonstrate that the MAML-based approach achieves better performance than traditional methods, validating its effectiveness for enhancing model adaptability and efficiency in few-shot AIOps scenarios, addressing the challenge related to limited labeled data and the necessity of frequent model updates.

Additionally, AIOps methodologies have found implications in other IT scenarios, intersecting notably with DevOps, DataOps, GitOps, and MLOps methodologies as explained in Mansour et al. (2024). These integrations promote automation, agility, and collaboration, helping enterprises reduce complexity and support continuous innovation (Diaz-De-Arcaya et al., 2023).

The integration of LLMs represents a particularly transformative advancement in the AIOps domain, specifically addressing tasks involving complex and unstructured data such as logs and incident reports (Vitui & Chen, 2025). Unlike traditional anomaly detection methods, which often require extensive manual feature engineering and retraining, LLMs have been trained on large amounts of cross-platform data and inherently possess robust capabilities for interpreting and extracting meaningful insights from natural language data (Vitui & Chen, 2025; Zhang, Jia et al., 2024). Their sophisticated NLP techniques allow them to efficiently analyze and understand context, semantics, and patterns within textual operational data, significantly streamlining anomaly detection and root cause analysis tasks (Vitui & Chen, 2025; Zhang, Jia

```
Jun 15 04:06:18 combo su(pam_unix)[21416]: session opened for user cyrus by (uid=0)
Jun 15 04:06:19 combo su(pam_unix)[21416]: session closed for user cyrus
Jun 15 04:06:20 combo logrotate: ALERT exited abnormally with [1]
Jun 15 04:12:42 combo su(pam_unix)[22644]: session opened for user news by (uid=0)
```

Fig. 7. Example log entry from var/log/messages.

et al., 2024). However, despite these advantages, LLMs still face notable limitations, including computational efficiency, constraints on cross-task adaptability and the need for careful model tuning and adaptation to specific operational contexts (Zhang, Jia et al., 2024). Nonetheless, their advanced NLP capabilities continue to offer substantial potential for enhancing the accuracy, interpretability, and adaptability of AIOps systems, making them a promising area for ongoing research and development (Vitui & Chen, 2025; Zhang, Jia et al., 2024). The importance of deploying these techniques in military contexts cannot be overstated. Military IT systems increasingly rely on autonomous platforms, secure communication infrastructures, and integrated cyber–physical operations that produce complex, mission-critical logs. These systems require not only accurate but also explainable and rapid anomaly detection methods to mitigate cyber threats, detect system failures, and maintain operational continuity under constrained conditions. Thus, the military sector presents both a high-need and high-impact setting for applying AIOps and LLM-based solutions.

4.2. LLMs in log anomaly detection

This section examines the application of Large Language Models (LLMs) in log anomaly detection. We discuss their capabilities, current use cases, and limitations, as well as how they compare to traditional anomaly detection methods.

Log anomaly detection refers to the process of automatically identifying unexpected patterns within log data generated by software systems (Guo et al., 2024). The problem is defined as a binary classification task, and the models are supposed to determine whether an input log is normal or abnormal (Guo et al., 2024). Logs are semi-structured or unstructured text generated by logging statements in source code (Hadadi et al., 2024; Mehrabi et al., 2024). They are used to monitor system performance and diagnose errors. An example log from /var/log/messages file on a Linux[13] server is included in Fig. 7.

As IT systems grow in complexity, generating large amounts of data that record the internal logic of systems and events, log anomaly detection has grown in importance. Effective anomaly detection within these logs is crucial for system reliability, security, and operational efficiency (Guan et al., 2024; Guo et al., 2024, 2023; Hadadi et al., 2024). The techniques used for log anomaly detection have evolved substantially, incorporating traditional ML, deep learning, and, most recently, leveraging LLMs to address the challenges of log data.

Logs typically contain diverse event types, unstructured messages that often contain noise (e.g., irrelevant information, typos, ambiguous expressions), which complicates the preprocessing for effective analysis (Su et al., 2024). Class imbalance is another significant challenge, as anomalous events occur less often than normal events. This imbalance can bias model training, where strategies such as minority class oversampling may be needed to ensure balanced learning (Guan et al., 2024). Furthermore, the availability of labeled datasets for supervised ML is limited in real-world scenarios, as anomalies are inherently rare and labeling is resource-intensive (Su et al., 2024). To address this challenge, log parsing has been proposed as a prerequisite for many log analytics tasks (Mehrabi et al., 2024). However, the challenge of log parsing is to distinguish automatically between static and dynamic tokens (Mehrabi et al., 2024). A possible solution could be the use of regular expressions (Mehrabi et al., 2024). Still, the problem is that log files may contain thousands of events, and any modifications to the system would require constant changes to the regular expressions (Mehrabi et al., 2024). Additionally, it is common for companies to have multiple types of log files, which complicates the development of regular expressions to parse the data (Mehrabi et al., 2024). Another challenge is the requirement for models to process logs in real-time. Logs are produced continuously, demanding anomaly detection models to balance accuracy with computational efficiency (No et al., 2024). Moreover, these logs frequently evolve as software gets updated. This needs models that can adapt quickly and generalize to new log patterns without extensive retraining (Hadadi et al., 2024).

Traditional methods for log anomaly detection often rely on statistical or rule-based approaches, such as Support Vector Machines (SVM), Isolation Forests (IF), Principal Component Analysis (PCA), SR, FFT, and PCI (Hadadi et al., 2024; Poenaru-Olaru et al., 2024). These methods require manual feature engineering, so they limit the scalability and generalization for new logs (Hadadi et al., 2024; Poenaru-Olaru et al., 2024). Additionally, there are deep-learning methods, including LSTMs and transformers, that automate feature extraction to capture the sequential nature of the log (Hadadi et al., 2024; Poenaru-Olaru et al., 2024). For example, LogBERT (Guo et al., 2021) is a self-supervised framework for log anomaly detection based on transformer architecture (BERT). The objective of LogBERT is to predict masked log keys in log sequences, where they are randomly masked and predicted, and hypersphere minimization to cluster the distribution of normal log sequences in the embedding space. The model is trained on normal logs and flags sequences with many incorrectly predicted log keys as anomalies. They use Drain (He et al., 2017) for parsing the log messages. Drain structures messages into a tree-based representation to facilitate online parsing, which is critical for real-time anomaly detection (He et al., 2017). Also, the authors do evaluations on three datasets (Hadoop Distributed File System (HDFS), BlueGene/L (BGL), and Thunderbird), and demonstrate that LogBERT outperforms other traditional methods such as PCA, IF, OCSVM, DeepLog,[14] and LogAnomaly[15] with higher F1-scores on the three datasets. Other evaluation metrics they use are precision and recall. Since automated log parsing is an essential part of this traditional pipeline to detect anomalies, Zhu et al. (2018) show a comprehensive evaluation of different automated parsing tools such as Drain (He et al., 2017), Spell, LogCluster, or MoLFI,[16] among others, across 16 log datasets from different domains, including distributed systems, operating systems, supercomputers, and server applications. They define log parsing as a step to convert raw logs into a structured

[13] More information: https://github.com/logpai/loghub/blob/master/Linux/README.md

[14] DeepLog is a deep learning-based model for anomaly detection in system logs, which uses LSTM networks to learn log sequence patterns under normal conditions and identify anomalies as deviations from those patterns. More information: https://github.com/Thijsvanede/DeepLog

[15] LogAnomaly is a deep learning-based framework for log anomaly detection that combines semantic and sequential modeling by using template embeddings and an LSTM network to identify abnormal log patterns. More information: https://www.ijcai.org/proceedings/2019/0658.pdf

[16] MoLFI (Multi-objective Log pattern Finder) is a search-based algorithm designed to discover log patterns from unstructured log messages using multi-objective optimization. It aims to balance quality and coverage. More information: https://github.com/SNTSVV/MoLFI

template. They base their findings on accuracy, robustness, and computational efficiency when determining anomalies within the log files, so future researchers can select an optimal parser based on their specific use case.

New strategies have emerged with the advancements of LLMs in anomaly detection through improved semantic understanding. For example, LogLLM (Guan et al., 2024) integrates BERT for extracting semantic vectors from log messages, and LLaMa[17] for classification of the log sequences. Additionally, Guan et al. (2024) introduce a projector for vector representation to achieve a cohesive semantic interpretation. Log messages are grouped into sequences based on session windows. Instead of relying on log parses such as Drain (He et al., 2017), the approach replaces specific objects in the log message with regular expressions. This technique is chosen for its simplicity and because existing parsers often struggle with out-of-vocabulary (OOV) words (Guan et al., 2024). Evaluation on four datasets (HDFS, BGL, Thunderbird, and Liberty) shows that on average LogLLM's F1-scores are 6.6% better than other existing methods such as DeepLog, LogAnomaly, PleLog, FastLogAD,[18] RAPID (No et al., 2024), and LogBERT (Guo et al., 2021). Other evaluation metrics they use are precision and recall. Similarly, LogFormer (Guo et al., 2024) employs a new attention mechanism and adapter-based tuning strategy to maintain semantic information across multiple log domains, enabling effective transfer learning. First, the pre-trained model learns the commonalities among different anomalies through a supervised classification task. Then, the parameters of the attention encoder are used in the tuning stage, where they prove the knowledge obtained from pre-training. In the preprocessing stage, Guo et al. (2024) also use Drain (He et al., 2017) to do log parsing of the sliding windows and select 80% of the messages for training and the remaining 20% for testing. Another approach is OWL (Guo et al., 2023), which shows the advances of LLMs in log anomaly detection by training a model on Operations and Maintenance (O&M) data. The authors employ a Homogeneous Markov Context Extension (HMCE) method to extend the context processing capabilities of LLMs. OWL uses a tunning technique called Low-Rank Adaptation (LoRA), which involves fine-tuning a small set of parameters in the model to reduce computational costs while maintaining performance. Then, it is evaluated on multiple benchmarks, including datasets from system architecture, application logs, infrastructure, and information security, showing higher performance in terms of F1-score when compared to other baseline models such as GPT-4, Qwen,[19] LLaMa, DeepLog, LogAnomaly, and LogRobust. (Xu & Ding, 2024) further provide a taxonomy for anomaly and OOD detection using LLMs and categorize their approaches into prompt-based and contrast-based methods. They describe prompt-based methods as creating constructed prompts that guide the LLM to directly output detection results using techniques such as role-play prompting, in-context learning, and CoT. Furthermore, they describe contrast-based methods as using multimodal LLMs to extract and compare embeddings for anomaly detection. Another novel method is LogFiT (Almodovar et al., 2024), which leverages a pre-trained BERT model fine-tuned on normal log data to detect anomalies. LogFiT eliminates the need for log parsing and instead uses a masked sentence prediction for self-supervised tuning. The authors state that log parsers could make inaccurate parsing due to misinterpretation of the semantic meaning of the log analysis and not handle OOV words well so which is why they propose this strategy. LogFiT detects anomalies based on the top-k token prediction accuracy, making it highly adaptable to different logs. Evaluations on HDFS, BGL, and Thunderbird datasets demonstrate that LogFiT outperformed baseline methods such as DeepLog and LogBERT (Guo et al., 2021) in terms of F1-score, precision, and recall. Lastly, LogPrompt (Liu et al., 2023) builds on these advancements by introducing a set of prompting strategies for interpretable online log analysis. These strategies include self-prompt, which leverages the intrinsic capabilities of the LLM to generate the output, chain-of-thought (CoT) prompt, which emphasizes a step-by-step reasoning process by guiding the LLM to address the task, and in-context prompt, which gives the LLM multiple examples to establish a contextual understanding, enhancing the interpretability of anomaly detection without requiring extensive in-domain training. The authors evaluate this model across multiple public datasets (HDFS, BGL, Linux,[20] Android,[21] Spirit) against other baseline models such as Drain (He et al., 2017), DeepLog, LogAnomaly, LogRobust, outperforming these models in terms of F1-scores, interpretability, and usability.

Another innovative methods that are growing nowadays are the ones that explicitly integrate external knowledge into the LLM for log anomaly detection. These examples are LogExpert (Wang et al., 2024) and RAPID (No et al., 2024). LogExpert leverages domain-specific knowledge from technical forums, such as Stack Overflow,[22] dynamically retrieving relevant knowledge to generate outputs. Its methodology fine-tunes LLMs on structured logs combined with domain-specific text information. LogExpert achieves strong performance of public datasets including HDFS, BGL, and Thunderbird, employing lexical and human-evaluated metrics to ensure accuracy and interpretability (Wang et al., 2024). In addition, RAPID (No et al., 2024) introduces a training-free, retrieval-based log anomaly detection method that uses LLMs. How RAPID works is that it categorizes logs based on system context and contrasts test logs against similar normal logs. They evaluate their experiments on benchmark datasets including BGL, Thunderbird, and HDFS, achieving an overall performance of 0.97 in F1-score. This model includes token-level information, which is the granular semantic details of individual log message tokens, and it is used to identify small deviations to detect anomalies. This strategy demonstrates robust performance in real-time scenarios without the need for domain-specific training.

However, despite these innovations, the use of LLMs for log analysis also poses challenges. Logs often contain private information, so using a proprietary LLM would force companies to send their data to a third party, with the risk of violating privacy regulations (Mehrabi et al., 2024). From a development standpoint, integrating third-party LLMs into existing log analysis workflows is also challenging (Mehrabi et al., 2024). Additionally, the substantial size of log files leads to high processing costs when using LLM-based approaches (Mehrabi et al., 2024).

Overall, log anomaly detection has evolved over the past years, from traditional, manual feature engineering ML methods to more advanced techniques employing deep learning and LLMs. These new advancements address complex challenges such as data representation, class imbalance, label scarcity, stream processing, and generalization

[17] LLaMA (Large Language Model Meta AI) is a family of open-source foundational language models developed by Meta, designed for efficient training and deployment in various natural language processing tasks. More information: https://ai.meta.com/llama/

[18] FasLogAD is a lightweight and efficient anomaly detection framework for log data that leverages statistical and semantic features to identify anomalies in real time, without relying on deep learning models. More information: https://arxiv.org/pdf/2404.08750

[19] Qwen is a series of large language models developed by Alibaba Cloud, designed for tasks such as text generation, reasoning, and code completion. The models are open-source and optimized for both general and domain-specific applications. More information: https://github.com/QwenLM

[20] The Linux dataset consists of logs from a production Linux-based system, often used for anomaly detection and root cause analysis. Retrieved from Loghub: https://github.com/logpai/loghub

[21] The Android dataset contains logs collected from an Android smartphone system, primarily used for studying mobile system reliability and crash analysis. Retrieved from Loghub: https://github.com/logpai/loghub

[22] Stack Overflow is a community-driven Q&A website for programmers where developers ask and answer questions on a variety of topics. More information: https://stackoverflow.com

to new logs. furthermore, as shown in the examples, LLMs enhance accuracy and interpretability when tested against traditional methods. Among these new approaches, the use of RAG is a promising technique that integrates external knowledge to improve context-awareness to further enhance anomaly detection.

A practical adoption decision requires trading off the improved semantic understanding of LLM-based solutions against their computational, financial, and operational costs. Computationally, lightweight statistical or reconstruction methods (e.g., SR, FFT, PCA) and tree- or ensemble-based algorithms (e.g., IF, SVM) have low inference latency and small memory footprints, making them suitable for high-throughput real-time pipelines (Poenaru-Olaru et al., 2024). By contrast, full-sized LLMs incur much higher memory and compute requirements at inference, increasing latency and requiring GPU/TPU resources or costly API usage (Brown et al., 2020; Mehrabi et al., 2024). Financially, LLM deployments may require cloud GPU instances or paid inference APIs, and RAG pipelines add vector-database and embedding-computation costs (e.g., FAISS indexing and re-ranking) that further increase total cost of ownership. Techniques such as adapter/LoRA tuning (OWL; Guo et al., 2023) and training-free retrieval methods (RAPID; No et al., 2024) can substantially lower tuning and training expenses, shifting costs toward inference and retrieval. Operational costs differ as well: conventional methods are easier to integrate, maintain, and explain to operators, whereas LLM/RAG systems add pipeline complexity (retrieval, prompt engineering, vector DB maintenance), pose privacy risks when using third-party models, and demand additional monitoring to detect model drift and hallucinations (Mehrabi et al., 2024; Ovadia et al., 2023).

Regardless of whether LLMs or traditional models are used for log anomaly detection, certain metrics, such as F1-score, precision, and recall, are commonly prioritized when evaluating model performance (see Table A1). These metrics are widely adopted because they capture key aspects of anomaly detection effectiveness: precision quantifies the proportion of correctly identified anomalies among all flagged events, recall measures the ability to detect all true anomalies, and the F1-score provides a balanced harmonic mean of the two (Almodovar et al., 2024). However, these metrics have limitations in the context of anomaly detection. For example, precision and recall are sensitive to class imbalance, which is common in anomaly detection tasks where anomalies are rare. A model may achieve high recall but at the expense of many false positives, potentially overwhelming IT operators, while high precision with low recall might miss critical anomalies. Moreover, the F1-score, by equally weighting precision and recall, may not reflect the operational priorities of specific domains, such as military systems, where missing an anomaly could have far more severe consequences than raising a false alert (Jeon et al., 2025). To address these limitations, future work could incorporate additional metrics, such as Matthews Correlation Coefficient (MCC) or area under the Precision-Recall curve (AUC-PR), and domain-specific cost functions that better reflect the criticality of anomalies in high-stakes environments.

4.3. RAG

This section explores Retrieval-Augmented Generation (RAG) as a technique to enhance the performance of LLMs in log anomaly detection. We explain the underlying mechanism of RAG, its advantages in domain adaptation and contextualization, and its relevance within the AIOps landscape.

LLMs show remarkable level of knowledge in various domains due to their pre-training datasets. However, there are limitations to this knowledge as it does not update and it is non-specific, meaning that it may lack expertise in specific domains (Ovadia et al., 2023). To solve this, an additional postprocessing step referred as knowledge injection is essential to add knowledge to the pre-trained model (Ovadia et al., 2023). There are two main frameworks for knowledge injection: fine-tuning and RAG (Ovadia et al., 2023). Fine-tuning is the process of adjusting a pre-trained model on a specific dataset to improve performance on that domain, and it can be classified into supervised, unsupervised, and reinforcement learning methods (Ovadia et al., 2023). On the other hand, RAG is an advanced method designed to improve the capabilities of LLMs by integrating external knowledge through the retrieval of databases using a semantic similarity calculation (Gao et al., 2023). It addresses limitations such as hallucinations, outdated information, immemorization, or reasoning failure from LLMs (Ovadia et al., 2023). RAG follows a process that includes indexing, retrieval, and generation (Gao et al., 2023). Indexing consists of cleaning and extracting the raw data and converting them into a uniform format, retrieval consists of prioritizing the fragments that contain the most similarity to the input query, retrieving them by calculating the similarity (e.g. cosine similarity) and using them as context in the prompt, and generation is the step where the input query and the selected documents are summarized into a prompt to which the LLM formulates a response (Gao et al., 2023). Although fine-tuning methods can improve LLM performance, Ovadia et al. (2023) show through different experiments that RAG outperforms this approach, especially when experiencing knowledge outside the original training set. The framework that Ovadia et al. (2023) use evaluates RAG against just the base model (Mistral-7B, Orca2-7B, and LLama2-7B), and unsupervised fine-tuning for the topics of anatomy, astronomy, college biology, college chemistry, and prehistory. Although fine-tuning improves results compared to the base model, it is not competitive to RAG, and it might be because RAG not only adds knowledge to the model but also incorporates context, which is a feature that fine-tuning lacks (Ovadia et al., 2023).

RAG employs different retrieval techniques classified as sparse or dense retrieval methods. Sparse retrieval methods, such as BM25 or TF-IDF, rely on lexical matching between the embeddings of the question and document chunks, whereas dense retrieval methods use neural embedding models for semantic searches (Gao et al., 2023). For example, Zhang, Jia et al. (2024) employ the Dense Passage Retrieval (DPR) framework that uses embedding models to generate dense vectors of queries. Other advanced methods for RAG include pre-retrieval query reformulation, where the focus is on optimizing the structure of the original query re-ranking based on relevant context retrieved, and adding metadata to improve the relevance of the retrieval (Gao et al., 2023). Fig. 8 shows a RAG interface pipeline from user input to response generation. The user submits a query, which the system may refine for clarity. Then it retrieves relevant information using hybrid (dense retrieval) or term-based search (sparse retrieval). Lastly, it filters and ranks the results, and injects them into the prompt so the LLM can generate a context-aware response.

In the specific context of log data, these retrieval techniques show different trade-offs. Sparse retrieval is effective for structured tokens in logs, such as identifiers, but struggles with semantic variation in log messages. Dense retrieval, by contrast, captures semantic similarity between log messages better, making it more suitable for heterogeneous or evolving systems, although it requires more computational resources and domain-specific tuning. Hybrid retrieval combines both approaches, balancing the precision of lexical matching with the semantic coverage of dense embeddings. This could be advantageous for logs, as they often contain a mixture of structured identifiers and unstructured descriptions, where neither sparse nor dense retrieval alone is sufficient.

Regarding evaluation metrics to validate the retrieval quality, Gao et al. (2023) identify metrics such as Mean Reciprocal Rank (MRR), Hit Rate (HR), and Normalized Discounted Cumulative Gain (NDCG). Common metrics for evaluating the quality of the answer include BLEU, ROUGE, F1 scores, exact match (EM), and precision (Gao et al., 2023). Additionally, frameworks such as RAGAS, ARES, and TruLens employ LLMs to also evaluate the quality scores (Gao et al., 2023).

The growing interest in RAG has led to its deployment across different domains and industries. Fan et al. (2024) offer a comprehensive taxonomy of RAG along with key applications divided into

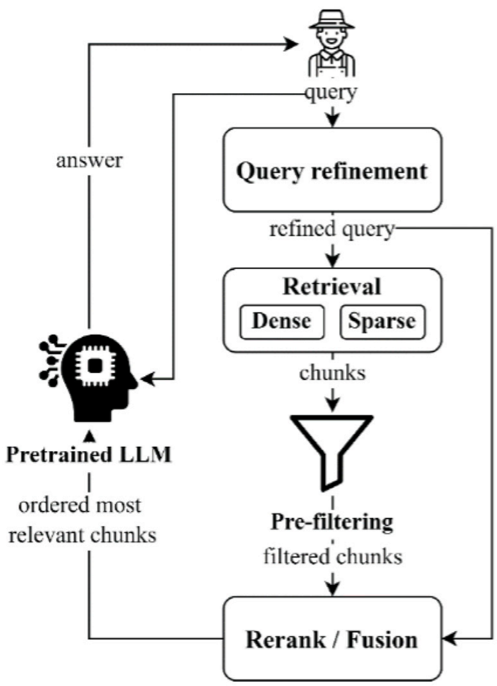

Fig. 8. Rag pipeline (Vizniuk et al., 2024).

three perspectives: NLP applications, downstream tasks, and domain-specific applications. NLP applications include Q&A systems, chatbots, and fact verification. In downstream tasks, RAG supports personalized recommender systems and software engineering workflows such as code generation, data preprocessing, text-to-SQL semantic parsing, and program repair. Beyond general-purpose use, domain-specific applications span fields such as AI for science, finance, healthcare, and education (Fan et al., 2024). Arslan et al. (2024) also support these observations with a large-scale survey that maps RAG applications across two principal categories: task-based classification and discipline-based classification. Under task-based classification, there are applications in the areas of Q&A, text generation and summarization, information retrieval and extraction, text analysis and processing, software development and maintenance, and decision-making (Arslan et al., 2024). Similarly, discipline-based classification includes domains such as medical/biomedical, finance, education, technology and software development, social and communication, and literature (Arslan et al., 2024).

While these taxonomies highlight the breadth of RAG's potential, other studies focus on its concrete implementation in specific domains, demonstrating how these general categories translate into real-world systems. In agriculture, Vizniuk et al. (2024) show how RAG can support decision-making scenarios that require expertise knowledge, such as crop cultivation, pest management strategies, or irrigation scheduling. The use of RAG in agriculture faces many challenges, such as the lack of domain-specific datasets (e.g., crop varieties, soil conditions, pest behavior, and climate variability), heterogeneous data sources, and ethical concerns around data privacy and security (Vizniuk et al., 2024). Vizniuk et al. (2024) also discuss a methodology that integrates external databases with regional agricultural knowledge that includes pest outbreak records, crop yield data, and weather patterns. The approach highlights the importance of expert validation alongside retrieved knowledge to minimize risks from potential inaccuracies. In industrial machine monitoring, Jeon et al. (2025) introduce ChatCNC, a conversational AI system framework that integrates LLMs to enable interactions with real-time Computer Numerical Control (CNC) machine data directly from operational databases using SQL queries. The methodology involves three agents: a Question Identifier that interprets questions from users, a Data Retriever that manages data preprocessing and connects to the real-time database, and a Response Generator that analyzes the retrieved information and gives responses to users. Jeon et al. (2025) evaluate ChatCNC by human raters providing scores ranging from 1 to 5 in different areas, such as user satisfaction with the bot, run time to process each question, and perceived helpfulness to operators. Regarding IT operations, the quality of maintenance varies depending on the operator's personal experience (Zhang, Jiang et al., 2024). Zhang, Jiang et al. (2024) propose the RAG4ITOps framework based on RAG to facilitate Q&A systems for IT operations and maintenance. The methodology is composed of two stages: (1) supervised fine-tuning of embedding models and data vectorization, and (2) online Q&A system process. After preprocessing the data, the first stage vectorizes the text chunks as embeddings and stores them in a vector database (FAISS[23]). The embedding models are fine-tuned using contrastive learning enhanced with negative sampling methods, such as Homogeneous In-Batch Negative Sampling (HIS) and Auxiliary Hard Negative Sampling (AHNS). The LLM is fine-tuned using RAG, where it retrieves the top-k relevant chunks based on the input query. In the second stage, the operators can ask questions. Then, the embedding model transforms the question into an embedding, and it is used to retrieve the most relevant context from the database. Finally, the LLM can answer the specific question by referring to all the contents in the input prompt. Zhang, Jiang et al. (2024) use datasets that include operational data (e.g., tool descriptions, operation examples, scripts, and system configurations) and maintenance data (error logs labeled by humans with solutions) provided by operators. The authors evaluate the framework using retrieval accuracy metrics and use GPT-4 as a scoring model to evaluate responses on a scale of 1 to 10. Similarly leveraging RAG to address operational queries, Chen et al. (2024) introduce the OMAI framework, designed as a Q&A system to assist operational staff within daily tasks specialized in the domain of high-energy physics. OMAI (Chen et al., 2024) uses the Xiwu[24] LLM, fine-tuned using a Helpdesk Q&A dataset by the Institute of High Energy Physics (IHEP). The framework retrieves external knowledge in the same way as RAG4ITOps (Zhang, Jiang et al., 2024). The evaluation of the framework employs metrics focused on domain-specific precision and operational reliability and shows that OMAI outperforms Vicuna-13B[25] and ChatGPT when handling tasks related to high-energy physics.

Overall, the evidence across the different studies, retrieval techniques, evaluation frameworks, and implementations proves that RAG has emerged as a robust solution to improve the knowledge limitations of pre-trained models. It injects external knowledge and enables models to reason with up-to-date information, and domain-specific context. Therefore, as the field continues to evolve, RAG not only stands out as a complementary alternative to fine-tuning, but also as a foundational architecture when building context-aware systems.

Despite promising results, the datasets used across many cited studies (e.g., HDFS, BGL, Thunderbird, Liberty) present limitations that challenge the generalizability of findings (Chen et al., 2024; Fan et al., 2024; Jeon et al., 2025; Zhang, Jiang et al., 2024). These datasets often focus on narrow environments such as distributed systems, supercomputers, or infrastructure logs and may lack diversity in terms of log structure, anomaly types, and operational contexts. For example,

[23] FAISS (Facebook AI Similarity Search) is a library developed by Facebook AI Research for efficient similarity search and clustering of dense vectors. More information: https://github.com/facebookresearch/faiss

[24] Xiwu is a foundational language model architecture that underpins the development of specialized systems for operational maintenance. More information: https://example.com/xiwu

[25] Vicuna-13B is an open-source chatbot model derived from Meta's LLaMA, fine-tuned on user-shared conversation data to enhance chat performance. More information: https://vicuna.lmsys.org/

BGL and HDFS logs follow consistent formatting and domain-specific patterns that may not reflect the variability found in logs from IoT, military, or hybrid cloud environments (No et al., 2024). Furthermore, labels in these datasets are typically human-annotated post hoc, which can introduce bias or inconsistency. These limitations are echoed in recent literature outside the AIOps domain, such as the work by Farooq et al. (2022), who emphasize that robust cyberattack detection requires diverse, representative datasets spanning multiple network types and intrusion modalities. Their approach underscores the need to evaluate AI-based anomaly detection methods across heterogeneous data sources to ensure real-world applicability. As such, future studies applying RAG and LLMs for log anomaly detection—especially in mission, critical domains, should validate across broader, multi-domain benchmarks to mitigate overfitting to curated datasets and enhance deployment readiness.

While LLM- and RAG-based approaches offer superior semantic understanding and adaptability, they also introduce notable trade-offs. Their high computational and memory demands increase inference latency and operational costs, which can be problematic for real-time anomaly detection (Brown et al., 2020). Traditional methods such as PCA or Isolation Forests are more lightweight and easier to deploy in resource-constrained settings (Poenaru-Olaru et al., 2024). LLMs may also hallucinate, producing plausible but incorrect outputs that risk false positives (Ji et al., 2023). Although RAG mitigates this by grounding responses in external knowledge, its performance depends heavily on retrieval quality and well-curated knowledge bases (Chen et al., 2024; Lewis et al., 2020). Moreover, these approaches add integration complexity, requiring maintenance of retrieval pipelines and fine-tuning, and can raise privacy concerns when logs are shared with third-party APIs, particularly in sensitive domains like the military (Jeon et al., 2025; Mehrabi et al., 2024). Hence, while LLMs and RAG mark a step forward in anomaly detection, their deployment should be carefully weighed against resource constraints, system complexity, and risk tolerance.

Building on the literature reviewed, we propose a taxonomy for classifying LLM- and RAG-based approaches for log anomaly detection (see Fig. 9). This taxonomy highlights three principal dimensions: (1) Model Architecture, (2) Knowledge Augmentation Strategy, and (3) Inference Objective.

- **Model Architecture:** LLM-based approaches for log anomaly detection can be broadly grouped into three categories: (a) *Transformer-based models*, which leverage pre-trained architectures such as BERT or LLaMA and fine-tune them for anomaly detection (e.g., LogLLM, LogFormer); (b) *Adapter-based or LoRA-tuned models*, which modify only a small subset of parameters to reduce computational overhead while maintaining performance (e.g., OWL); and (c) *Prompt-based or zero-shot methods*, which use prompting strategies – such as self-prompting, in-context learning, and chain-of-thought – to guide the model toward anomaly detection without task-specific retraining (e.g., LogPrompt).
- **Knowledge Augmentation Strategy:** Knowledge integration is a key differentiator between purely LLM-based and RAG-based approaches. We classify these strategies into (a) *No External Knowledge*, where the model relies solely on its pre-trained weights (e.g., LogFiT); (b) *Implicit Knowledge Injection*, where domain adaptation is achieved via fine-tuning or adapters on domain-specific logs; and (c) *Explicit Knowledge Augmentation*, which uses retrieval-based methods to dynamically incorporate external sources such as documentation, technical forums, or knowledge bases (e.g., LogExpert, RAPID). RAG methods can be further subdivided into *sparse retrieval*, *dense retrieval*, and *hybrid retrieval* approaches.
- **Inference Objective:** Finally, LLM/RAG systems differ in how they frame the anomaly detection task. We distinguish between (a) *Classification-based methods*, which output a binary label indicating normal or anomalous behavior; (b) *Ranking-based methods*, which score log events according to anomaly likelihood and allow operators to set thresholds; and (c) *Explanatory methods*, which not only flag anomalies but also provide natural-language rationales or root-cause explanations (e.g., LogPrompt's interpretability focus).

In line with recent recommendations in the AIOps literature, we also emphasize the importance of explainability for deep learning models applied to log anomaly detection. Ante-hoc methods, such as model-driven deep unrolling, offer a way to embed interpretability into the learning process itself by unfolding iterative optimization algorithms into trainable neural network architectures, enabling operators to understand how individual network layers relate to classical inference steps. Post-hoc approaches complement this by providing visual explanations of model predictions, for instance through saliency maps, attention heatmaps, or token-level importance scores that highlight which log segments contributed most to the anomaly classification. Together, these methods enhance trust and transparency in AIOps pipelines, which is crucial for operational decision-making, root-cause analysis, and compliance with regulations in safety-critical domains. By combining these dimensions, researchers and practitioners can better map the design space of LLM/RAG-based log anomaly detection systems, identify complementary techniques, and systematically compare their performance across datasets and operational constraints.

4.4. Theoretical framework and future applications

This section presents a theoretical framework for leveraging LLMs and RAG in log anomaly detection within AIOps environments. We outline the key components and interactions of the framework and propose directions for future research to validate and extend its practical applicability. Given the growing technological sophistication of modern defense systems, future applications in the military domain are particularly pressing. Log anomaly detection tools that integrate LLMs and RAG are well-positioned to address operational needs in areas such as mission planning, battlefield communication, cyber defense, and unmanned vehicle diagnostics—domains where the cost of undetected anomalies may be catastrophic.

This research aimed to conduct a comprehensive SLR on the advancements of AIOps to use LLMs for log anomaly detection within the military industry. However, few studies in the existing literature focus on LLM-based approaches for log analysis within the military context. For our research, we identified two works that, although not directly targeting log anomaly detection, do explore broader aspects of LLMs and AI in military scenarios. Loevenich et al. (2024) describe a framework for integrating autonomous cyber defense agents into a simulated NATO-inspired network environment. The study integrates Multi-Agent Reinforcement Learning (MARL), LLMs, and rule-based systems to automate cybersecurity tasks such as monitoring, detection, and mitigation to increase the capabilities of cybersecurity professionals (Loevenich et al., 2024). The document includes a segmented network architecture in which blue agents are deployed to defend multiple security zones against red adversaries operating within an unstructured network. In parallel to this development, Cui and Gao (2024) examine the explainability of generative AI in military contexts. Cui and Gao (2024) analyze how the "black-box" problem in generative AI conflicts with decision-making in military contexts and propose a three-stage framework with input, interpretation, and feedback as the main components. The input module allows users to provide input to the model, the interpretation module aims to explain the decision-making of the model, and the feedback module allows users to provide feedback and guidance to the system, enabling future improvements of the model (Cui & Gao, 2024). In addition, Cui and Gao (2024) outline potential military applications of generative AI, such as reconnaissance and intelligence

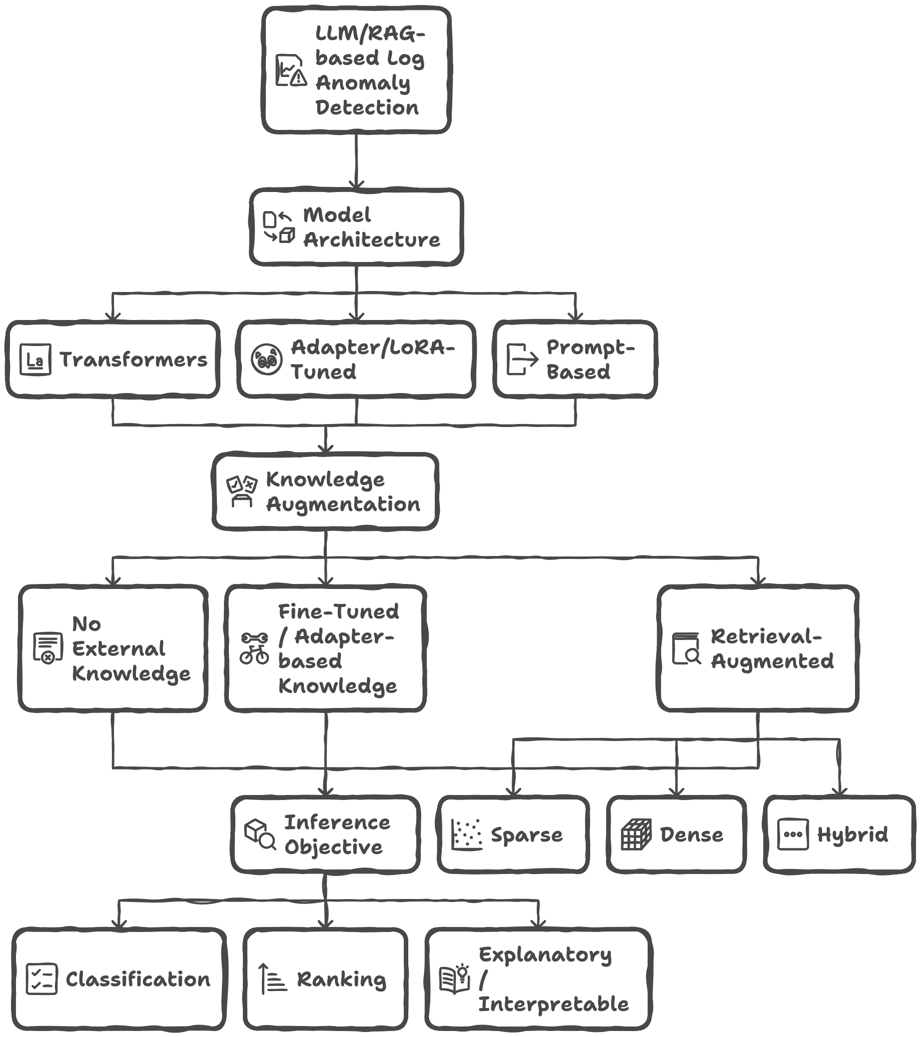

Fig. 9. Taxonomy of LLM- and RAG-based log anomaly detection methods.

gathering, operational command, and operational control, and provide two examples highlighting the principle of explainability: an approach that uses LLMs with knowledge graphs to improve military search tasks, and a target-strike selection mechanism that combines decision trees with generative AI (Cui & Gao, 2024). Despite the limited research on log anomaly detection with LLMs in military settings, the insights of these applications highlight valuable opportunities. The frameworks discussed offer a foundational base to bridge this gap in the literature.

Building upon these papers, future work on the use of LLMs for log anomaly detection within the AIOps field can focus on several directions that aim to improve applicability, efficiency, and specialization. Real-time anomaly detection, RAG-tailored log datasets, and this new application in the military domain represent key areas of exploration.

Most studies analyzed so far operate in an offline mode, which can limit the timeliness of anomaly detection. For example, if the failure perception time window is set to 10 s (an event is triggered every 10 s), the model must infer results within 1 s to promptly notify users in case of anomalies, and currently, no LLM-based work has adequately tackled this problem (Zhang, Jia et al., 2024). Future research can investigate real-time anomaly detection pipelines, where logs are continuously ingested, parsed, and analyzed with minimal latency to optimize the computational efficiency of LLMs for AIOps. One potential direction of research could explore methods that combine lightweight feature extraction modules (such as Drain-based parsing) to structure logs fast, followed by a fine-tuned or prompt-engineered LLM that is capable of handling the log streams. Another promising direction is applying RAG to log datasets rather than typical Q&A systems. RAG improves the LLM's generative capabilities with external context retrieved from documents, enabling more domain-specific insights. To the best of our knowledge, systematic RAG-based frameworks for log analysis are still rare, as most existing RAG systems target Q&A tasks outside of log analysis. Future implementations could store log templates in vector databases, allowing the system to retrieve similar entries when a new log template arrives and then consolidate the retrieved content within the LLM's prompt. This could enable the model to generate context-aware anomaly classifications, potentially reducing hallucinations and improving interpretability. Finally, research on applying LLMs explicitly to military log analytics remains scarce. Future studies could investigate how domain-specific knowledge bases may be retrieved via RAG to give LLMs specialized context in the military industry.

We propose a framework (see Fig. 10) to guide researchers in implementing a robust log anomaly detection system using LLMs with RAG. The framework has three main stages: definition of the project, database construction, and anomaly detection, mapped onto the CRISP-ML[26] methodology, which is designed to provide a structured approach to ML projects from initial concept to final deployment. The first stage, the definition of the project, aligns with the understanding of the business. Researchers formulate a scientific problem that concerns the

[26] More information: https://ml-ops.org/content/crisp-ml

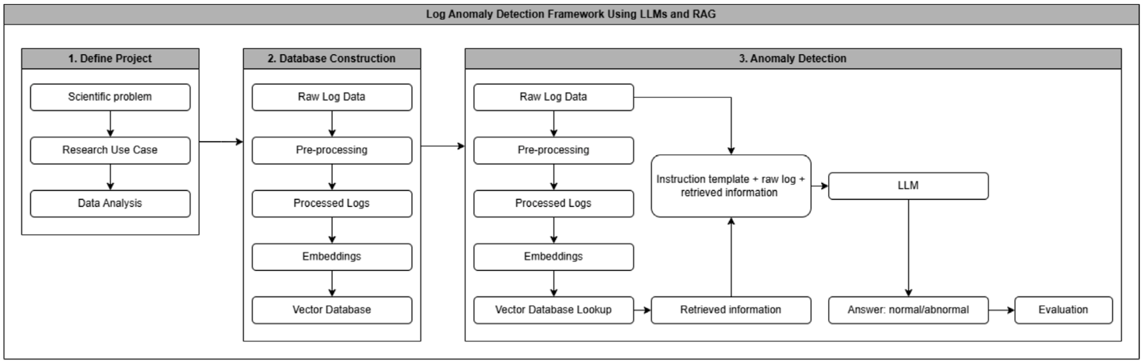

Fig. 10. Framework.

limited capacity of traditional rule-based or statistical systems to detect log anomalies in complex and high-volume data environments, as traditional systems struggle to adapt to evolving system behaviors or fail to capture the semantic context of log messages. The research use case investigates how LLMs can semantically interpret log messages, integrate context via RAG, and deliver more accurate and interpretable log anomaly classifications. Specifically, it addresses the needs of environments such as enterprise IT systems in the military domain, where the detection of abnormal events is critical for IT operations. Furthermore, it identifies key stakeholders, including IT operators, system administrators, and maintainers. This stage also involves data collection, preparation, exploration, and visualization analysis to better understand the nature of the datasets. The second stage of the framework establishes the vector database. After data preparation, researchers read an anomaly log file in chunks and parse the raw logs into a structured representation to generate embeddings from the processed logs using pre-trained embedding models. These embeddings capture semantic patterns and represent the logs in a vector space. Researchers store the resulting vectors in a vector database to enable efficient lookups during anomaly detection. This infrastructure allows the system to extract context from previously processed logs, supporting a RAG-based interface. The vector database makes the anomaly detection process both context-aware and scalable to evolving requirements. Finally, the anomaly detection stage in the framework aligns with the modeling and evaluation phases of CRISP-ML. New log messages pass through a similar preprocessing pipeline to that used for historical logs. The system then leverages the vector database to retrieve any relevant context (similar logs, documented incidents, known anomaly patterns) and feeds this context along with the incoming log entries and an instruction template into the LLM. The LLM receives these merged inputs and returns a classification decision, labeling the logs as normal or abnormal. This stage incorporates an evaluation methodology to validate the LLM output.

To sum up, real-time anomaly detection, specialized RAG for logs, and applications in the military domain can mark the next phase of innovation in LLM-based log anomaly detection as an integral part of future AIOps strategies. The framework proposed follows CRISP-ML principles by beginning with a clear foundation of the research problem and use case, proceeding through data analysis, establishing a modeling infrastructure through a vector database, and concluding with an anomaly detection phase that includes an LLM and the evaluation of the outputs. This framework ensures rigorous technical development and continual improvement for building an anomaly detection system using LLMs and RAG.

5. Conclusion

This paper presented a SLR on how AIOps techniques can benefit from LLMs to improve log anomaly detection in modern IT systems. First, we described the core tasks of AIOPs (data preprocessing, failure perception, root cause analysis, and auto remediation) and explain the challenges of logs due to their complexity and changing nature. Next, we compared traditional anomaly detection methods with advanced LLM-based approaches, including fine-tuning, prompt engineering, and RAG. These approaches showed more accurate and interpretable results in both experimental and enterprise settings.

Regarding the knowledge questions proposed in the introduction, we identified several answers. The primary challenges in implementing AIOps include dealing with concept drift, obtaining high-quality labeled datasets, and adapting to dynamic environments, while benefits involve increased reliability, scalability of IT infrastructures, fast detection and diagnosis, and reduction of manual effort. Common techniques used for anomaly detection within AIOps are traditional ML methods, deep learning models (LSTM, and transformers), and more recently, advanced use of LLMs. The main evaluation metrics in log anomaly detection using LLMS encompass precision, recall, F1-score, and AUROC. Lastly, the use of RAG substantially improves LLMs capabilities by introducing context from external sources, thereby reducing hallucinations or outdated information, and enabling models to reason with domain-specific context.

Although LLM-driven solutions show strong potential, organizations must address key challenges such as data privacy concerns, computational efficiency of LLMs, and limited application in critical domains such as military. Our study highlights domain-specific training and the use of RAG as promising strategies to solve these challenges. By using LLMs to particular operational contexts, AIOps can accelerate anomaly detection and be used as an external tool to help IT operators, system administrators, and maintainers. Focusing on the military domain is especially relevant given the increasing reliance of defense operations on autonomous systems, remote sensing technologies, and AI-driven command structures—all of which generate massive and sensitive log data. Ensuring resilience and operational integrity in such environments requires anomaly detection systems that go beyond conventional methods, leveraging LLMs and RAG to provide context-sensitive and timely responses. Although this study emphasizes military applications, the proposed approaches and insights are broadly generalizable to other high-stakes sectors, including healthcare, finance, and industrial systems, where reliability and rapid anomaly detection are equally vital.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Table A1.

Table A1
Table reporting all the articles examined to conduct the research and highlighting their main features ("/" = None).

Topic	Author	Settings	Main purpose	Data-driven techniques	Metrics for evaluation
AIOps	Zhang, Jia et al. (2024)	AIops for failure management	Comprehensive survey of AIops technology for failure management in the LLM era. Includes detailed definition of AIops tasks for failure management, data sources for AIops, and LLM-based approaches adopted for AIops	Foundational model, fine-tuning approach, embedding-based approach, prompt-based approach	–
Anomaly detection	Su et al. (2024)	LLMs for forecasting and anomaly detection	Systematic literature review that examines the application of LLMs in forecasting and anomaly detection, highlighting the current state of research, challenges, and future directions.	Prompt-based, fine-tuning, zero-shot, one-shot, and few-shot	Precision, accuracy, recall, F1-score, AUROC
AI in military	Cui and Gao (2024)	Generative AI in military contexts	Explore the development of explainable generative AI in military contexts, followed by a framework to achieve explainability in generative AI.	Knowledge graphs	–
AI in military	Loevenich et al. (2024)	Military networks	Development and training of robust autonomous cyber defense agents within military networks	Multi-Agent Reinforcement Learning, LLMs, and a rule-based system	–
AIOps	Dang et al. (2019)	AIops challenges and research innovations	Real-world challenges in building AIOps solutions based on our practice and experience in Microsoft. Then they propose a roadmap of AIOps related research directions	–	–
AIOps	Vitui and Chen (2025)	AIops in large-scale enterprise environments (RedHat Openshift/K8s)	Propose an LLM-powered AIOps approach to automate workflows, enhance efficiency, and support decision-making in modern IT Operations. Integrate LLMs (e.g., GPT/Claude/Mistral) with a predictive ML tool (MLASP) to tackle capacity planning and other ITOM tasks.	Use LLMs for reasoning (ReAct framework), predictive ML for planning (MLASP), and RAG for text/document retrieval, plus other tools for real-time data queries	Compliance with instructions (how effectively the model follows user prompts), accuracy (ratio of correctly predicted instances to the total number of instances), latency and throughput (time required for the model to generate responses), cost (token usage)
AIOps	Prasad and Rich (2019)	Market Guide aimed at enterprise-scale IT operations	Provides a market overview of how AIops platforms use big data and ML to optimize IT operations.	–	–
Anomlay detection	Liu et al. (2023)	LLMs for log analysis (parsers + anomaly detection)	Propose LopPrompt, a prompt-enegeering approach that leverages LLMs to perform online log parsing and anomaly detection, with no in-domain training. Emphasizes interpretability via chain-of-thought and in-context prompts to generate explanations along with predictions.	LLMs (ChatGPT and Vicona 13B) for zero-shot/few-shot parsing and anomaly detection. Advanced prompt strategies: self-prompt, chain-of-thought, and in-context learning	F1-score for log parsing, session-level F1-score for anomaly detection, and human evaluation for interpretability.
Anomaly detection	Hadadi et al. (2024)	LLMs for anomaly detection on unstable logs	Compare fine-tuned GPT-3 vs traditional ML/DL baseline models vs GPT-4 prompt-engineering for detecting anomalies in logs that change over time. Investigate how LLM pre-training can help mitigate data scarcity.	Fine-tuned LLM for sequence classification, and prompt-based approach (zero/few-shot) for anomaly detection. They use multiple baseline models: PCA, IM, LogCLuster, DeepLog, LogAnomally, PLELog, LogRobust, CNN, NeuralLog	Precision, recall, F!-score (session-level anomaly detection. Also, they use Fisher's Exact Test for statistical significance of differences
Anomaly detection	Xu and Ding (2024)	Anomaly and OOD detection with LLMs	Comprehensive survey of how LLMs can be leveraged to detect anomalies or out-of-distribution samples in diverse data. Proposes a taxonomy separating "LLMs for detection" vs "LLMs for generation".	Prompt-based approaches (in-context prompting, chain-of-thought), Contrasting-based detection, and LLM-generated data augmentations and explanations	AUROC, AUPR, F1-score, and false positive rate.
AIOps	Duan et al. (2024)	Meta-Learning for Efficient Adaptation in Few-Shot AIOps Scenarios	Propose MAML/KAD, a meta-learning-based anomaly detector for AIOps tasks, focusing on how few-shot learning addresses scarcity of failure data.	Few-shot learning methods (MAML/ANIL) for anomaly detection and classification in AIOps.	Accuracy, F1-score, zero-shot adaptation metrics (evaluated on log and KPI datasets) comparing baseline vs MAML.
Anomaly detection	Guo et al. (2024)	LLms for Log anomaly detection	Propose a transformer-based framework for log anomaly detection (LogFormer) to improve generalization across different domains. The model is pre-trained on source domain to obtain a general knowledge of log data. Then, the knowledge is transferred to the target domain	Drain for log parsing, pre-trained sentence-bert for feature extraction of the template sequence, and adapter-based tuning to leverage the knowledge obtained during the pre-training.	Baseline models: SVM, Deeplog, LogAnomaly, LogRobust, PLELog, 2 variants of LogFormer, and ChatGPT. For evaluation they use precision, recall, and F1-score. For practical evaluation the model was successfully applied to a cloud servicice of a company.
Anomaly detection	Mehrabi et al. (2024)	Comptact fine-tunned LLM in log parsing	Discuss the challenges of data privacy, cost, and tool integration regarding LLMs. They explore the viability of supervised fine-tuning of an open-source compact LLM for log parsing.	Fine-tuning Mistral-7B via LoRA for log parsing (zero/few-shot vs GPT-4).	Metric-based approach: accuracy (message level accuracy and Levenshtein edit distance) and F1-score. LLM-based approach: use of GPT-4 as a log parser evaluator to rate the event templates form 0 to 5.
RAG	Zhang, Jia et al. (2024)	Retrieval-augmented Q&A for IT operations	Proposes RAG4ITOps, a comprehensive pipeline that collects enterprise-private docs and logs, fine-tunes an embedding model using contrastive learning with custom negative sampling, fine-tunes a generative LLM (Qwen-14b) with retrieval-augmented fine-tuning for domain Q&A, and supports an online Q&A workflow	Fine-tuned dense passage retriever (LoRA on a BGE-M3 backbone) plus retrieval-augmented instruction tuning of Qwen LLM; negative sampling (in-batch/hard negatives), chunked corpora, and custom instruction templates for knowledge acquisition and troubleshooting Q&A	Top-k retrieval accuracy, single-score mode (select first one model, and generate answers based on given questions, and then use GPT-4 as a scoring model to evaluate responses from 1 to 10), and pairwise-score (both models generate answers to identical questions using the same reference chunks and then a scoring model asses which models's reponses are better

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Table A1 (*continued*).

RAG	Gao et al. (2023)	Survey paper on RAG	Provide a comprehensive review of the evolution and state-of-the-art methods in RAG, outlining its development from early Naive RAG to Advanced and Modular RAG. The paper categorizes various techniques, discusses the underlying retrieval, generation, and augmentation processes, and evaluates the practical integration of RAG within large language models.	Vector-based retrieval: Use sparse embeddings to represent and retrieve text chunks. Indexing and query optimization: Methods such as chunking strategies, metadata enrichment, query expansion, rewriting, and routing. Iterative and adaptive retrieval: Approaches that refine and enhance context through multiple passes. Fine-tuning: Tailoring both embedding models and generation models on domain-specific data.	The paper reviews various metrics used to assess RAG systems across two main dimensions. Retrieval: Hit Rate, Mean Reciprocal Rank, Normalized Discounted Cumulative Gain, cosine similarity. Generation: Accuracy, Exact Match, F1 score, BLEU, ROUGE.
Anomaly detection	Guo et al. (2023)	LLM for IT operations	Introduce OWL, a specialized large language model for IT operations, detailing its custom dataset, novel long-context extension (HMCE), and parameter-efficient tuning (Mixture-of-Adapter), and demonstrating its superior performance on the Owl-Bench and other IT tasks.	Data Augmentation: Owl-Instruct constructed via expert seed data and GPT-4–aided augmentation with strict quality control. Long-Context Extension: HMCE method uses a homogeneous Markov chain for extending input length. Efficient Tuning: Mixture-of-Adapter strategy for task-specific fine-tuning	Evaluated on Owl-Bench (Q&A and multiple-choice), long-context inference (perplexity), and downstream IT tasks like log anomaly detection and log parsing using metrics such as F1-score, precision, recall, and RandIndex
Anomaly detection	Zhu et al. (2018)	Automated log parser survey	Present a comprehensive evaluation study of automated log parsing methods by benchmarking 13 log parsers on 16 real-world log datasets, and releasing an open-source toolkit (logparser).	Reviews a variety of data-driven approaches from frequent pattern mining, clustering, heuristics (iterative partitioning, longest common subsequence, parsing trees, and evolutionary algorithms) to automatically extract event templates from unstructured log messages.	Accuracy: Parsing accuracy (PA) that is defined as the ratio of correctly parsed log messages. Robustness: consistency of performance across different log types and volumes. Efficiency: processing time.
Anomaly detection	He et al. (2017)	Online log parsing for web service management	The authors propose Drain, an online log parsing method that employs a fixed depth parse tree with specially designed rules to efficiently and accurately parse raw log messages in a streaming manner, enabling timely log analysis and effective anomaly detection in large-scale web service environments	Fixed-depth parse tree, token similarity (simSeq) to match log messages with log events.	Parsing Accuracy: Based on how well-parsed log messages match the ground-truth log events (uses F1-score). Efficiency: running time improvements. Effectiveness: demonstrated through a case study on anomaly detection
RAG	Chen et al. (2024)	LLMs for operational maintenance	Introduces OMAI, a specialized LLM designed to enhance IT operations at IHEP by reducing staff workload through a Q&A system. It also uses fine-tuning and RAG to deliver more accurate responses	Fine-tuning, RAG integration. The model is built on Xiwu.	Model evaluated via internal benchmarks showing an effective response rate exceeding 90% on common operational queries, indicating superior performance than ChatGPT
Anomaly detection	Guan et al. (2024)	LLMs for log anomaly detection	Introduce LogLLM, a framework that leverages both transformer-based (BERT) and decoder-based (Llama) LLMs to detect anomalies in logs by extracting semantic vectors, aligning their representations via a projector, and classifying the log sequences through a novel three-stage training.	Preprocessing: regular expressions instead of log parsers. Semantic extraction through BERT. Vector alignment is done through a projector that aligns the embedding spaces of BERT and Llama. Classification is done with Llama, and training requires a three-stage fine-tuning process.	Precision, recall, and F1-score
AIOps	Mansour et al. (2024)	Integration od new operational methodologies in IT MLEs in Germany	This article prmarily serves as a literature review and analyzes the adoption, benefits, challenges, and integration strategies of DevOps, AIOps, DataOps, GitOps, and MLOps in medium-to-large IT enterprises in Germany, and offers a holistic roadmap to improve operational efficienct and competitiveness.	–	–
AIOps	Diaz-De-Arcaya et al. (2023)	MLOps and AIOps challenges and opportunities	The paper provides a systematic survey addressing challenges and opportunities associated with MLOps and AIOps. The goal is to clarify key issues and present frameworks that support the development of AI/ML solutions.	–	–
AIOps	Notaro et al. (2021)	Addressing failure management in IT operations through AIOps	The article provides a survey of AIOps approaches for failure management by categorizing contributions into five areas (failure prevention, failure detection, failure prediction, root-cause analysis, and remediation), and discusses future trends.	Reviews a wide range of AI and ML methods, including statistical models (linear/logistic regression, Bayesian networks), machine learning classifiers (SVM, decision trees, random forests), clustering, neural networks (LSTM, deep learning), and Markov models.	It summarizes metrics reported in other studies such as accuracy, precision, recall, F1-score, false-positive/negative rates, MSE.
AIOps	Poenaru-Olaru et al. (2024)	AIOps for anomaly detection in IT operations.	The article investigates the impact of concept drift on anomaly detection models and compares different maintenance strategies. It examines blind (periodic) versus informed (drift-triggered) retraining, and full-history versus sliding window retraining approaches to determine how best to adapt AIOps solutions to real-world data changes.	Some state-of-art anomaly detection models(FFT, spectral residual, LSTM), and integrate a concept drift detector (FEDD) to trigger model updates.	F1-score, precision, and recall.

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Table A1 (*continued*).

Anomaly detection	No et al. (2024)	Address challenges in IT operations by targeting real-world log anomaly detection in computer systems	This paper proposes RAPID; a retrieval-based log anomaly detection method. The goal is to leverage pre-trained LLMs and token-level information to detect anomalies in log files in real time without the need of specific training.	BERT as pre-trained models. Anomaly detection is reformulated as a retrieval problem where each query is compared to the normal documents using the cosine similarity. A KNN algorithm is employed to select a 'core set' from the document database.	F1-score and AUROC. Also, the authors use latency to validate real-time applicability.
Anomaly detection	Wang et al. (2024)	Automated log anomaly detection to aid site reliability engineers in troubleshooting large-scale software systems.	The articles proposes a novel framework called LogExpert. It generates recommended steps to deal with anomalous logs. It integrates the powerful text comprehension of LLMs with domain-specific knowledge extracted from Stack Overflow	Log recognizer: It uses a BERT classification model to automatically identify software logs from noisy data. Extractive summarization: the authors use ASSORT to summarize long posts to reduce the number of tokens. Resolution generator: they use Faiss vector database to retrieve the top-k similar logs and the LLM to generate the steps.	Accuracy, precision, recall, F1-score, BLEU-4, ROUGE-L, and human evaluations.
RAG	Arslan et al. (2024)	Survey paper in the academic research of RAG	It reviews the key architectures, training strategies, application areas, and current limitations of implementing RAG.	–	–
RAG	Fan et al. (2024)	Survey paper on RA-LLMs in the context of digital transformation and NLP research.	This paper is a systematic literature review about existing research on RA-LLMs, examining the architectures, applications, and training strategies. The goal is to show how RAG can augment LLMs to deal with issues such as hallucinations, internal limitations, and outdated knowledge.	–	–
RAG	Vizniuk et al. (2024)	RA-LLMs integration into DSSs for agrotechnical monotoring	The goal of this article is to analyze the state-of-art research on developing and integrating LLMs with RAG for agricultural decision making. It highlights functional limitations of LLMs such as explainability, or biases in DSS applications and outlines interesting direction to improve efficiency and sustainability in the future.	–	–
Anomaly detection	Guo et al. (2021)	Anomaly detection for online computer systems	The paper proposes LogBERT, a self-supervised framework for log anomaly detection. It aims to learn normal log sequence patterns via two training tasks (masked log key prediction and volume of hypersphere minimization) and then detect anomalies as deviations from these learned patterns	BERT-based transformer. Masked Log Key Prediction (MLKP) where randomly masked log keys are predicted, and Volume of Hypersphere Minimization (VHM) encourages normal log sequences to cluster together in the embedding space.	Namely precision, recall, and F1 score
Anomaly detection	Almodovar et al. (2024)	System log anomaly detection	The paper introduces LogFiT, which is a log anomaly detector that uses pre-trained models such as BERT. It is designed to work on raw data without the need of log parsing. The model learns the semantic and sequential patterns of normal logs, so it flags deviations as anomalies.	Transformer-based LLM with self-supervised training. The model relies on top-k token prediction accuracy to decide whether the new log deviates from the normal behavior learned	Precision, recall, F1-score. The experiments are conducted on three datasets (HDFS, BGL, Thunderbird)
RAG	Jeon et al. (2025)	Conversational Q&A in manufacturing environments using RAG	Introduces ChatCNC, a framework that uses LLM-based multi-agent collaboration to enable natural language queries and real-time data retrieval from CNC machines. This allows operators to access and interpret data without specialized technical skills.	RAG to retrieve real-time CNC data via SQL queries. Use of pre-trained models (GPT-4, LLaMa3-8B, Mistral 7B) to classify questions, generate queries, and produce responses. Use of prompt engineering (persona, context, exemplar) to tailor the LLMs	Response accuracy: categorizing responses based on correctness. Failure mode: categorizing errors. Human evaluation: measure user satisfaction regarding helpfulness and runtime performance. They rate them using a 1-5 scale.
RAG	Ovadia et al. (2023)	LLM Knowledge injection	Compare two ways of adding new or specialized knowledge to pre trained LLms (Minstral7B, Llama2-7B, Orca2-7B): unsupervised fine-tuning and RAG. Also, it investigates how repeated paraphrasing helps LLMs learn new information	Unsupervised fine-tuning (continuation of pretraining with domain text), Retrieval Augmented Generation (retrieving context from a vector database), Data Augmentation (paraphrases of the same fact)	Accuracy on multiple-choice tasks, relative accuracy gain vs baseline, and training loss curves

Data availability

Data will be made available on request.

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